CS360 Nano Quiz - 25.04.21(Mon)

1. Please choose a WRONG explanation about secondary storage.

(1) Database systems typically store data on secondary storage devices such as HDD and SSD, since main memory is volatile.

(2) A single disk block typically consists of one of more consecutive sectors on the disk.

(3) Disk access time consists of seek time, rotational latency, and transfer time.

(4) Transfer time is calculated as (size of blocks / transfer rate). So, transfer time varies depending on the transfer rate of the secondary storage type (e.g., HDD, SSD).

(5) The rule of thumb that sequential I/O is much faster than random I/O holds true only for HDDs, not for SSDs.

2. Please choose a WRONG explanation about RAID and buffering technique

(1) We assume that R is the time to read one block into the buffer, and P is the time to process one block in the buffer. Let n be the total number of blocks. Then, for database applica9ons using the double buffering technique, the total elapsed time of compute-intensive application usually becomes (R + nP).

(2) RAID 0 is fast and cheap, but not reliable.

(3) RAID 5 or 6 is the best option in terms of both reliability and speed.

(4) RAID 10 is the best option in terms of both speed and reliability, but its cost is high.

3. Please choose a WRONG explanation about records and blocks.

(1) A record consists of consecutive bytes in disk block(s).

(2) Although records are stored in disk, they are eventually manipulated in main memory, and so, their layout should be optimized for efficient access in main memory.

(3) Both a record and a block begin with fixed-length metadata, referred to as a header.

(4) If a record has 12-byte header and a single string field of 6 bytes. Then, it is laid out as 18 bytes within a disk block.

(5) The header of a record contains a pointer to the schema, since the record itself does not store schema information about the relation to which it belongs.

CS360 Nano Quiz - 25.04.23(Wed)

1. Please choose a WRONG explanation about database address space.

(1) The memory address of a block is the virtual-memory address of its first byte.

(2) The physical address of a data object includes device ID for the disk, cylinder number, track number, and so on.

(3) Logical address is shorter than physical address.

(4) The slotted page format is one of possible combinations of logical and physical addresses.

(5) <u>Although using the slotted page format, an entry of logical and physical addresses of a record MUST exist in the global map table.</u>

(6) In the slotted page format, we can handle the deletion of the record easily by leaving a tombstone in the record's entry in the offset table.

2. Please choose a WRONG explanation about pointer swizzling.

(1) Memory address is much more efficient than database address.

(2) The translation table translates from database address to memory address.

(3) Accessing data objects in main memory through the translation table uses indirection and so is time consuming.

(4) Pointer swizzling allows us to avoid the cost of translating repeatedly from DB address to Mem address.

(5) <u>When the block is loaded into memory, automatic swizzling leaves all pointers unswizzled.</u>

(6) A database pointer contains a single bit indicating whether the current pointer is either database address or memory address.

3. Please choose a WRONG explanation about pinned blocks.

(1) A pinned block means the block that cannot be written back to disk safely.

(2) If Block B1 has a swizzled pointer to a data item in Block B2, then moving B2 back to disk and reusing its main memory buffer makes the pointer dangling.

(3) To write a pinned block back to disk, we need to unswizzle any pointers in the block and unpin the block by unswizzling any pointers to it.

(4) To unpin a block, we can use a linked list of pointers referring to data items in the block.

(5) <u>To create the above linked list in main memory, we need to allocate a separate memory space.</u>

CS360 Nano Quiz - 25.04.28(Mon)

1. Please choose a WRONG explanation about variable-length records and record modification.

(1) For the records with repeating fields, we can use an alternative method that puts the variable-length portion on a separate block.

(2) For storing the records that do not have a fixed schema like JSON and XML, we can use a sequence of tagged fields

(3) A single BLOB is usually stored on a sequence of blocks (i.e., spanned record).

(4) <u>A tombstone is not permanent; it can be re-used by garbage collection at some time before the entire DB is reconstructed.</u>

2. Please choose a WRONG explanation about index.

(1) An index is for speeding up the queries that specify values for one or more attributes.

(2) A single data file can have one or more index files.

(3) The number index entries of a dense index is the same with the number of records of the corresponding data file.

(4) <u>A sparse index can be used for any data file.</u>

(5) The second and higher-level index must be sparse

3. Please choose a WRONG explanation about secondary index.

(1) A secondary index is built on a non- primary key attribute.

(2) <u>Every level of a secondary index must be dense.</u>

(3) A secondary index may be useful in some cases where two relations are stored in a clustered file, and the secondary index is built on the attribute used in queries.

(4) A bucket file can reduce the space overhead of a secondary index.

(5) We can find the query result efficiently by intersecting two sets of pointers in bucket files.

CS360 Nano Quiz - 25.04.30(Wed)

1. Please choose a WRONG explanation about B-tree.

(1) The blocks of B-tree are organized into a balanced tree.

(2) B-tree is a disk-based index, and so each node of B-tree is a disk block.

(3) A leaf node must have at least 2 pointers pointing to data records, when n = 3.

(4) When n=3, a single interior node can have up to four pointers for B-tree blocks.

(5) <u>The root node also must be at least half full.</u>

2. Please choose a WRONG explanation about the performance of B-trees.

(1) For the same number of records, the depth of a B-tree is much smaller than that of a binary search tree due to its higher fan-out.

(2) Inserting a new key may increase the depth of a B-tree index (i.e., the number of levels may grow from n to n+1).

(3) Insertion or deletion may change the keys in the root node.

(4) <u>If a depth of B-tree is 5, and its root node is buffered In main memory, a lookup operation typically requires 6 disk I/O’s.</u>

(5) If a depth of B-tree is 5, and its root node is NOT buffered in main memory, an insertion operation typically requires 7 disk I/O’s.

CS360 Nano Quiz - 25.05.07(Wed)

1. Please choose a WRONG explanation about hash tables.

(1) The number of buckets of a static hash table is not changed.

(2) Both extensible hashing and linear hashing belong to dynamic hashing.

(3) A lookup operation in a static hash table always requires exactly one disk I/O.

(4) In extensible hashing, each nub value (i.e., j value) is always equal to or smaller than the i value.

(5) When i = 10 in extensible hashing, the bucket array can have up to 1024 entries.

(6) To use i = 10 in extensible hashing, the hash function must generate hash values of at least 10 bits.

2. Please choose a WRONG explanation about linear hash tables.

(1) The number of buckets in linear hash tables grows more slowly than the number of buckets in extensible hash tables.

(2) The average number of records per bucket MUST not exceed a give threshold (load factor).

(3) The high-order bits of a hash value are used to place records.

(4) Sometimes, an overflow block is created.

3. Please choose a WRONG explanation about multidimensional data space.

(1) Applications using multidimensional data space include GIS, IC design, MRI, and computer games.

(2) The "Where-am-I" query is one of the typical queries of GIS.

(3) Processing range queries on multidimensional data spaces using B-trees is not feasible.

(4) Processing GIS queries using one-dimensional indexes is usually less-efficient than processing them with multi-dimensional indexes.

(5) Multidimensional indexes are categorized into hash-table-like approach and tree-like approach.

4. Please choose a WRONG explanation about grid files.

(1) A grid file belongs to hash structures.

(2) The numbers of grid lines must be the same across all dimensions.

(3) Spacing between adjacent grid lines are usually non-uniform (i.e., varying).

(4) Some buckets may be empty.

(5) The insertion operation may add a new grid line, which again increases the number of buckets.

CS360 Nano Quiz - 25.05.12(Mon)

1. Please choose a WRONG explanation about partitioned hasing.

(1) A naive hashing method cannot compute h(a,b) if either a or b is not given.

(2) Partitioned hashing divides a total of k bits among n attributes, where n is equal to the number of hash functions.

(3) The i-th hash function h_i() produces k_i bits, and K_1 + K_2+... +k_n= k.

(4) The bucket for a tuple of (v_1, v_2, ..., v_n) is determined as h_1(v_1)h_2(v_2)... h_n(v_n).

(5) <u>Partitioned hashing is useful for nearest-neighbor queries or range queries.</u>

2. Please choose a WRONG explanation about kd-tree.

(1) Interior nodes generalize the binary search tree to multidimensional data.

(2) Leaf nodes are data blocks.

(3) Insertion operation may create a new interior node.

(4) <u>Partial-match queries always explore exactly one child at each node.</u>

(5) Range queries may need to explore both children at certain nodes.

(6) For adapting kd-trees to secondary storage, we need to pack multiple interior nodes into a single block for performance.

3. Please choose a WRONG explanation about R-tree.

(1) It captures the spirit of B-tree for multidimensional data.

(2) <u>Subregions within a region never overlap.</u>

(3) Subregions of a region usually do not cover the entire region.

(4) To represent a region or subregion, it uses an MBR.

(5) When inserting a new data object, if there is no subregion containing the object, one of the existing subregions is expanded to include it.

CS360 Nano Quiz - 25.05.14(Wed)

1. Please choose a WRONG explanation about bitmap index.

(1) <u>In run-length encoding, a run means a sequence of 1's followed by a 0.</u>

(2) The size of an uncompressed bitmap index can be n^2 (n: the number of records).

(3) It can answer range queries very efficiently by using both bitwise AND and bitwise OR.

(4) We can reduce the size of a bitmap index by using compression techniques like run-length encoding.

2. Please choose a WRONG explanation about the operations of bitmap index.

(1) Bitwise AND / OR operations can be performed without uncompressing the entire bit-vectors.

(2) A bitmap index can be built by thinking of each bit-vector as a record of (key, bit-vector).

(3) To find data records by using a bitmap index, we need to add a record ID column to the data file.

(4) Record IDs assigned for use with a bitmap index must not change after they are assigned.

(5) <u>Bit-vectors in a bitmap index must remain fixed after creation.</u>

CS360 Nano Quiz - 25.05.19(Mon)

1. Please choose a WRONG explanation about query processing.

(1) Query processor roughly consists of two parts: query compilation and query execution.
(2) Query compilation is done in three steps: parsing query, selecting logical query plan, and selecting physical query plan.
(3) To choose the best query plan among candidates, we need to estimate their costs.
(4) <u>Only one physical query plan can be generated from the best logical query plan.</u>
(5) Parsing step converts a character string of SQL statement into a tree structure.

2. Please choose a WRONG explanation about scan operators.

(1) There are two kinds of approaches: iteration (pipelining) and materialization.
(2) An iterator consists of three operations: Open(), GetNext(), and Close().
(3) Materialization means the result of an operator is produced entirely and stored on disk (or in main memory).
(4) There are three major scan operators: table scan, index scan, and sort scan.
(5) <u>The sort scan operator is only used when the ORDER BY clause is specified in a SQL query.</u>

3. Please choose a WRONG explanation about the cost model.

(1) The number of disk I/O's is used as a measure, for a disk-based DBMS.
(2) The cost can be changed depending on some parameters such as the number of buffers (M), the size of input relations, and the distribution of data.
(3) V(R, [a_1, a_2]) means the number of distinct values in a pair of columns a_1 and a_2 in a relation R.
(4) <u>The cost of the table scan is always equal to B(R), i.e., the number of blocks of a relation R.</u>
(5) The cost of the index scan is at least B(R), if the entire R is scanned.

CS360 Nano Quiz - 25.05.21(Wed)

1. Please choose a WRONG explanation about 1-pass algorithms.

(1) Duplicate elimination can be performed in one pass as long as B(delta (R)) <= M.

(2) For grouping operation, the hash(or search) key of each group entry is the value of the grouping attributes.

(3) For bag union, we need to build a hash table.

(4) Set union can be performed in one pass as long as min (B(R), B(S)) <= M.

(5) For bag intersection, the hash table store not only each distinct value, but also its count.

2. Please choose a WRONG explanation about 1-pass algorithms.

(1) Nested-loop joins are considered belong to one-and-a-half pass algorithms.

(2) Nested-loop joins can be used for relations of any size.

(3) Nested-loop joins is usually faster than other join algorithms.

(4) Block-based nested-loop join between S (outer) and R (inner) is faster than block-based nested-loop join between R (outer) and S (inner).

(5) The cost of block-based nested-loop join can be approximated as B(R)B(S) / M, regardless of the join order.

3. Please choose a WRONG explanation about sort-based two-pass algorithms.

(1) Sorting operation for a relation R can be done in two passes, when B(R) <= M^2.

(2) The 2nd pass of 2PMMS merges at most M-1 sorted sublists.

(3) Sort-based join may be slower than nested-loop join, in terms of disk I/ O, when data sizes is small.

(4) Sort-merge-join between relations R and S has the cost of 3B(R) + 3B(S).

(5) Improved sort-based join (i.e., sort-merge-join) is better than the basic sort-based join in terms of both disk I/O's and the max sizes of relations that can be processed.

CS360 Nano Quiz - 25.05.26(Mon)

1. Please choose a WRONG explanation about hash-based two-pass algorithms.

(1) The cost of hash-based duplicate elimination for R is 3B(R).

(2) Pass 1 of the hash-based join uses the join attribute as the hash key.

(3) Pass 1 of the hash-based join partitions both S and P into M-1 buckets.

(4) Pass 2 of the hash-based join builds a hash table for a larger chunk.

(5) The cost of hash-based join between R and S is 3B(R) + 3B(S).

2. Please choose a WRONG explanation index-based two-pass algorithms.

(1) In a clustering index, tuples appear on roughly as few blocks as can hold them, in terms of indexing attribute(s).

(2) A clustered relation always has a clustering index.

(3) A non-clustered relation CANNOT have an actual clustering index.

(4) Index-based join may be slower than hash-based join or sort-based join, although it utilizes an index.

(5) If both R and S have their own B-tree indexes, and the join is performed using the search keys of those indexes, then we can perform zig-zag join for R and S.